Prober-Generator Resonance Spectroscopy: Hierarchical Signature Coherence and Pure Cloze Model DNA Attribution Across Multi-Domain Discourse

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Abstract

Conventional machine text forensics treats provenance analysis as a binary thresholding problem driven by surface stylometrics (such as punctuation densities or lexical n-grams) or token-level log-likelihood curvature. However, surface features are trivially erased by prompt conditioning, and applying a single uncalibrated decision threshold across heterogeneous knowledge domains introduces severe calibration distortion. In this work, we introduce **Prober-Generator Resonance Spectroscopy**, a threshold-independent framework that situates generative provenance within a formal four-tier **Hierarchical Taxonomy of LLM Signature Coherence**. Rather than imposing a binary decision cutoff or extracting surface stylometric artifacts, we excise contiguous multi-sentence voids ($k \in \{2, 3, 4\}$) from candidate documents and measure how strongly an external frozen prober (`nemotron-3-super-120b`) reconstructs the excised propositional trajectory via permutation-invariant Kuhn-Munkres bipartite matching and context-anchor delexicalization. Evaluating a newly curated $N = 600$ **Multi-Slice Pure Cloze Forensic Benchmark** spanning seven provenance classes (Meta LLaMA-3.3, DeepSeek-V3/Qwen-2.5, Google Gemini-2.5, OpenAI GPT-4o, Anthropic Claude-3.5, NVIDIA Nemotron-3, and Human academic controls) across four cognitive slices (Abstract Foundations, Ethical Conundrums, Strategic Decisions, and Day-to-Day Applied prose), we uncover a distinct spectrum of **discursive determinism**. Under zero surface stylometrics, frontier model families separate sharply from the orthogonal human cognitive floor ($\mu_{\text{Human}} = 24.96\%$), led by Meta ($\mu = 49.30\%$), DeepSeek/Qwen ($\mu = 45.65\%$, raw cloze mean 44.37%), Google ($\mu = 44.35\%$), OpenAI ($\mu = 43.07\%$), and Anthropic ($\mu = 37.11\%$). Crucially, domain-slice analysis reveals that RLHF moral and safety alignment induces peak macro-discursive determinism in *Ethical Conundrums* (**75.76% AUROC**, **94.91% AUPRC**, +21.23% separation margin), followed by *Strategic Decisions* (**70.67% AUROC**, **93.48% AUPRC**). Finally, we demonstrate that prober alignment measures the generator's intrinsic discursive entropy rather than naive self-family recognition, establishing a rigorous mathematical and empirical foundation for continuous generative text spectroscopy.

1 Introduction

As foundational language models proliferate across scientific, legal, and enterprise ecosystems, characterizing the structural signatures that distinguish model families from human cognition—and from one another—has become a central problem in machine learning provenance (Mitchell et al., 2023; Bao et al., 2024; Hans et al., 2024). Historically, automated text forensics has relied upon two narrow methodological paradigms: (i) *surface-distributional stylometry*, which counts orthographic tokens, punctuation heuristics (such as em-dash or colon frequencies), and n-gram distributions (**?**), and (ii) *token-level probability curvature*, which evaluates local log-likelihood gradients under discrete token perturbations (Mitchell et al., 2023; Bao et al., 2024).

While computationally convenient, both paradigms fail to capture the deeper structural invariants of generative language models. Surface stylometric markers are ephemeral artifacts of post-training formatting

templates that vanish under simple prompt instructions or light editing. Meanwhile, token-level likelihood estimators conflate local lexical predictability with generative origin, degrading under syntactic paraphrasing (Krishna et al., 2023; Sadasivan et al., 2023) and suffering from severe calibration drift when a fixed decision threshold (τ) or sigmoid inflection center (c_0) calibrated on one domain is applied to dense technical or philosophical prose (Liang et al., 2023).

In this paper, we depart from binary thresholding and surface stylometrics to formulate **Prober-Generator Resonance Spectroscopy**. We posit that large language models trained via autoregressive pretraining and Reinforcement Learning from Human Feedback (RLHF) operate within structured, low-entropy macro-discursive manifolds. When a contiguous multi-sentence informational bridge ($k \in \{2, 3, 4\}$ sentences) is excised from the interior of a document, an external prober model conditioned on the surrounding context attempts to infill the missing propositions. By measuring the semantic and relational predicate alignment between the ground-truth excised span and the prober's counterfactual reconstruction, we directly quantify how strongly the document's deductive trajectory resonates with the prober's generative prior.

Core Contributions of This Work:

1. **Hierarchical Taxonomy of LLM Signature Coherence:** We formalize a four-tier ontological hierarchy classifying generative signatures from fragile surface stylometry (Level 1) and token curvature (Level 2) to rhetorical schema invariance (Level 3.1) and *Latent Attractor Basins & Infilling Topology* (Level 3.2, our formal class).
2. **Epistemics of Uncalibrated Thresholding vs. Continuous Spectroscopy:** We prove why evaluating binary False Positive Rates (FPR) on human text under a fixed sigmoidal inflection center ($c_0 = 0.70$) and cutoff (τ) across uncalibrated domains is methodologically invalid. We replace binary thresholding with continuous resonance spectroscopy ($\mathcal{R}(D; P_m)$), threshold-independent AUROC/AUPRC, and orthogonal cognitive baseline separation ($\Delta\mu$).
3. **Pure Cloze Engine V3 (Zero Surface Stylometrics):** We isolate pure semantic infilling by stripping all surface punctuation, character, and lexical frequency rules. We combine maximum-weight bipartite matching via the Kuhn-Munkres (Hungarian) algorithm (Kuhn, 1955) with context-anchor delexicalization, Cross-Span Minimum gating ($\min_i S_i$), and novel relational predicate congruence.
4. **The $N = 600$ Multi-Slice Forensic Attribution Benchmark:** We construct and evaluate a balanced $N = 600$ benchmark across four distinct cognitive domains ($n = 150$ each: Abstract Foundations, Ethical Conundrums, Strategic Decisions, Day-to-Day Applied) and seven provenance classes (Meta, DeepSeek/Qwen, Google, OpenAI, Anthropic, NVIDIA, and Human literature).
5. **Discovery of RLHF Alignment Determinism & Discursive Entropy:** We demonstrate that moral and safety RLHF alignment compresses generative variance in *Ethical Conundrums*, driving zero-shot pure cloze separability to **75.76% AUROC** (**94.91% AUPRC**, +21.23% resonance margin), with Google reaching 51.33% and Meta reaching 50.14% resonance against a 21.70% human floor. Furthermore, we show that prober responsiveness reflects intrinsic generator determinism rather than naive self-architecture affinity.

2 Hierarchical Taxonomy of Generative Signature Coherence

To establish a rigorous theoretical foundation for model attribution, we classify machine text signatures into a hierarchical taxonomy ordered by structural depth and invariance to surface perturbation (Table 1):

- **Level 1: Surface-Distributional Stylometry (Orthographic & Lexical):** Encompasses token frequency distributions, Zipfian tail anomalies, punctuation densities (e.g., em-dashes, colons, Oxford commas), and parse-tree readability indices (**?**). Because Level 1 features operate on discrete surface symbols, they possess near-zero invariance under system-prompt styling or basic human copy-editing.
- **Level 2: Token-Level Likelihood Curvature (Statistical Perplexity):** Evaluates local log-probability gradients $\nabla_w \log p_\theta(w_t \mid w_{<t})$ under single-token mask perturbations (Mitchell et al.,

Table 1: Hierarchical Classification of Generative Text Signatures and Provenance Paradigms.

Taxonomic Level	Signature Class	Representative Signals	Surface Invariance	Primary Attribution Target
Level 1	Surface Stylometry	Punctuation, n-grams, parse depth	None (Trivially spoofed)	Formatting / Prompt template
Level 2	Likelihood Curvature	Log-rank curvature, perplexity ratio	Low (Temperature sensitive)	Token-level decoder likelihood
Level 3.1	Rhetorical Invariance	Dialectical hedging, transition arcs	Moderate	RLHF conversational framing
Level 3.2 (Ours)	**Infilling Topology**	**Multi-sentence bipartite cloze resonance**	**High (Delexicalized propositions)**	**Macro-discursive model DNA**

2023; Bao et al., 2024) or paired-observer perplexity ratios (Hans et al., 2024). Level 2 features capture local syntactic smoothness but remain highly sensitive to decoding temperature, top-p sampling, and clause-level paraphrasing (Sadasivan et al., 2023).

- **Level 3: Structural Semantic Manifold & Discourse Resonance:** Captures macro-propositional coherence across multi-sentence spans, invariant to lexical synonym substitution. Level 3 bifurcates into two distinct subclasses:
  1. *Subclass 3.1 (Rhetorical & Dialectical Schema Invariance):* Characterizes macroscopic transitions, such as the canonical RLHF dialectical arc (*Thesis* $\rightarrow$ *Antithesis* $\rightarrow$ *Balanced Synthesis*) and hedging preambles.
  2. *Subclass 3.2 (Latent Attractor Basins & Infilling Topology — Model DNA):* Evaluates whether an excised multi-sentence void S_{void} can be counterfactually reconstructed by an external frozen prober P_m conditioned strictly on surrounding anchors $(S_{\text{prefix}}, S_{\text{suffix}})$.

3 Methodology: Pure Cloze Engine V3 & Resonance Spectroscopy

3.1 Continuous Prober-Generator Resonance Spectroscopy

Let a document $D = (s_1, s_2, \ldots, s_N)$ consist of N ordered sentences. We select a discourse interior centroid t and excise a contiguous multi-sentence void of horizon $k \in \{2, 3, 4\}$:

$$S_{\text{void}}^{(k)} = (s_t, s_{t+1}, \ldots, s_{t+k-1}), \quad S_{\text{ctx}}^{(k)} = S_{\text{prefix}} \cup S_{\text{suffix}} \tag{1}$$

where $S_{\text{prefix}} = (s_1, \ldots, s_{t-1})$ and $S_{\text{suffix}} = (s_{t+k}, \ldots, s_N)$. Given a library of frozen prober models $\mathcal{P} = \{P_1, \ldots, P_M\}$, prober P_m is prompted zero-shot to generate k infilling sentences $\hat{S}^{(k,m)} = (\hat{s}_1, \ldots, \hat{s}_k)$ bridging S_{prefix} and S_{suffix}.

Theorem 1 (Permutation Invariance via Kuhn-Munkres Bipartite Matching)**.** *Let $G = (S_{void}^{(k)}, \hat{S}^{(k,m)}, E)$ be a complete weighted bipartite graph with pairwise sentence congruence weights $W_{i,j} = \Phi(s_{t+i-1}, \hat{s}_j) \in [0, 100]$. The maximum-weight bipartite matching over the symmetric group $\mathcal{S}_k$:*

$$\pi^* = \arg\max_{\pi \in \mathcal{S}_k} \sum_{i=1}^{k} \Phi\left(s_{t+i-1}, \hat{s}_{\pi(i)}\right), \quad \mathcal{C}_k(D; P_m) = \frac{1}{k} \sum_{i=1}^{k} \Phi\left(s_{t+i-1}, \hat{s}_{\pi^*(i)}\right) \tag{2}$$

is strictly invariant to arbitrary clause permutations $\rho, \sigma \in \mathcal{S}_k$ of either ground-truth or infilled sentences, and executes in $\mathcal{O}(k^3) \leq 64$ operations (< 0.1 ms).

Proof Sketch. For any bijections $\rho, \sigma \in \mathcal{S}_k$, the composite mapping $\psi = \sigma \circ \pi \circ \rho^{-1}$ is a group automorphism of $\mathcal{S}_k$, preserving the global maximum over all $k!$ assignments (full proof in Supplementary Section A). □

3.2 Context-Anchor Delexicalization & Sigmoidal Dynamic Gating

To prevent shared topical vocabulary in $S_{\text{ctx}}^{(k)}$ from inflating semantic similarity in specialized domains, we define the context anchor vocabulary $\mathcal{V}_{\text{ctx}} = \{w \in S_{\text{prefix}} \cup S_{\text{suffix}}\}$ and project out context-shared noun stems prior to computing dense representation similarity $C_{\text{dense}}(s_{\text{delex}}, \hat{s}_{\text{delex}})$ (via `all-MiniLM-L6-v2`) and novel

relational predicate overlap $P_{\text{novel}}(s, \hat{s})$. Pairwise sentence congruence $\Phi(s, \hat{s})$ is governed by a continuous sigmoidal gate:

$$w_{\text{dense}}(C) = w_{\text{base}} + \frac{w_{\text{boost}} - w_{\text{base}}}{1 + \exp\left(-\kappa \cdot (C - c_0)\right)}, \quad \Phi(s, \hat{s}) = (1 - w_{\text{dense}}) P_{\text{novel}}(s, \hat{s}) + w_{\text{dense}} C_{\text{dense}}(s_{\text{delex}}, \hat{s}_{\text{delex}}) \tag{3}$$

where $w_{\text{base}} = 0.20$, $w_{\text{boost}} = 0.40$, $\kappa = 15.0$, and $c_0 = 0.70$. Across void scales $k \in \{2, 3, 4\}$, we compute the multi-scale mean $\bar{\mathcal{C}} = \frac{1}{3}(p_2 + p_3 + p_4)$, the Cross-Span Minimum Gate $S_{\min} = \min_i \Phi(s_i, \hat{s}_{\pi^*(i)})$, and syntactic predicate boost β_{pred} to yield the composite V3 Resonance score $\mathcal{R}_{\text{V3}}(D; P_m) \in [0, 100]$.

3.3 Epistemic Critique of Uncalibrated Human FPR vs. Resonance Spectroscopy

A critical methodological contribution of our framework is eliminating the misapplication of fixed binary False Positive Rate (FPR) cutoffs across uncalibrated probing domains:

Proposition 2 (Calibration Drift Under Fixed Inflection Center c_0)**.** *Let c_0 denote the sigmoidal gating center and τ denote a scalar decision boundary calibrated on a reference baseline corpus $\mathcal{D}_{ref}$ (where human baseline density satisfies $\mathbb{E}_{\mathcal{D}_{ref}}[C_{dense}] = \mu_0$). When transferred zero-shot to a formal academic domain $\mathcal{D}_{formal}$ with constrained technical nomenclature ($\mathbb{E}_{\mathcal{D}_{formal}}[C_{dense}] = \mu_0 + \Delta_{domain}$, $\Delta_{domain} > 0$), the effective gating shift $\tilde{c}_0 = c_0 - \Delta_{domain}$ induces a systematic upward shift in baseline scores, rendering any uncalibrated scalar cutoff τ statistically uninformative.*

Consequently, measuring a hard binary "Human FPR" at a fixed, uncalibrated τ across diverse cognitive slices (such as formal epistemology vs. casual workplace memos) conflates domain lexical density with classification error. In **Prober-Generator Resonance Spectroscopy**, we do not apply a brittle binary threshold. Instead, we evaluate:

1. **Threshold-Independent Separability (AUROC and AUPRC):** Integrating ranking separability across all possible operating thresholds $\tau \in [0, 100]$.
2. **Orthogonal Human Floor vs. Family Resonance Margin ($\Delta\mu$):** Because authentic human prose does not originate from a transformer probability manifold, human texts establish an unentangled cognitive baseline floor ($\mu_{\text{Human}} = 24.96\%$), against which we measure each model family's continuous resonance elevation $\Delta\mu_f = \mu_f - \mu_{\text{Human}}$.

4 The $N = 600$ Multi-Slice Pure Cloze Forensic Benchmark

4.1 Experimental Architecture and Slice Design

We construct a balanced, multi-domain forensic benchmark of $N = 600$ documents (100–450 words, ≥ 6 sentences per document), partitioned equally into four cognitive slices ($n = 150$ per slice: 128 synthetic outputs across six frontier model families and 22 authentic human academic controls):

- **Slice 1: Abstract Foundations ($n = 150$):** Epistemology, philosophy of mind, mathematical logic, and theoretical physics.
- **Slice 2: Ethical Conundrums ($n = 150$):** Bioethics, autonomous triage dilemmas, algorithmic governance, and multi-stakeholder moral conflicts.
- **Slice 3: Strategic Decisions ($n = 150$):** Game-theoretic equilibria, macroeconomic policy, supply-chain risk mitigation, and institutional strategy.
- **Slice 4: Day-to-Day Applied ($n = 150$):** Operational workflows, software post-mortems, executive communications, and procedural guides.

All 600 documents were evaluated zero-shot via Pure Cloze Engine V3 using `nvidia/nemotron-3-super-120b-a12b` as the frozen external infilling prober across three simultaneous void horizons ($k \in \{2, 3, 4\}$). Zero surface stylometric rules were permitted.

Table 2: Threshold-Independent Separability and Continuous Resonance Margins Across the Four Cognitive Slices of the $N = 600$ Pure Cloze Forensic Benchmark (Evaluated via Frozen Nemotron-3 120B Prober).

Cognitive Domain Slice	Total (N)	AI / Human	AI Resonance ($\mu_{AI} \pm \sigma$)	Human Floor ($\mu_{Hu} \pm \sigma$)	Resonance Margin ($\Delta\mu$)	Threshold-Free AUROC	Threshold-Free AUPRC
Slice 1: Abstract Foundations	150	128 / 22	36.37% ± 22.67%	24.90% ± 16.90%	+11.48%	64.58%	92.12%
Slice 2: Ethical Conundrums	**150**	**128 / 22**	**42.93% ± 22.89%**	**21.70% ± 17.27%**	**+21.23%**	**75.76%**	**94.91%**
Slice 3: Strategic Decisions	**150**	**128 / 22**	**40.69% ± 22.56%**	**24.31% ± 19.33%**	**+16.38%**	**70.67%**	**93.48%**
Slice 4: Day-to-Day Applied	150	128 / 22	39.32% ± 22.91%	28.93% ± 15.40%	+10.40%	62.93%	91.99%
Combined $N = 600$ Forensic Ledger	**600**	**512 / 88**	**39.83% ± 22.88%**	**24.96% ± 17.47%**	**+14.87%**	**68.76%**	**93.21%**

Table 3: Cross-Architecture Model DNA Resonance Spectrum, Pairwise Separability vs. Human Floor ($N = 88$), and Multi-Scale Sub-Component Decomposition across the $N = 600$ Benchmark.

Model Family / Provenance Class	N	V3 Resonance ($\mu \pm \sigma$)	Margin ($\Delta\mu$)	Family AUROC vs. Hu	Cloze Mean ($\bar{\mathcal{C}}$)	Scale p_2	Scale p_3	Scale p_4	Min Span ($S_{\min}$)	Novel Pred (P_{novel})
Meta (LLaMA 3.3 70B Instruct)	84	**49.30% ± 25.42%**	**+24.34%**	77.00%	32.78%	31.81%	33.13%	33.41%	21.91%	22.97%
DeepSeek / Qwen (V3 / Qwen 2.5)	84	**45.65% ± 17.76%**	**+20.69%**	**80.68%**	**44.37%**	**41.81%**	**44.88%**	**46.41%**	**27.91%**	**35.41%**
Google (Gemini 2.5 Flash Lite)	88	**44.35% ± 24.06%**	**+19.39%**	72.97%	32.31%	31.24%	32.34%	33.33%	21.27%	21.60%
OpenAI (GPT-4o / GPT-4o-mini)	88	**43.07% ± 23.62%**	**+18.11%**	72.41%	34.70%	33.49%	34.96%	35.66%	22.62%	25.72%
Anthropic (Claude 3.5 Sonnet)	84	**37.11% ± 18.86%**	**+12.15%**	68.11%	32.50%	31.68%	32.57%	33.25%	22.97%	21.67%
Top-4 Canonical Families Pooled	**344**	**45.56% ± 23.04%**	**+20.60%**	**75.69%**	**35.98%**	**34.45%**	**36.26%**	**37.12%**	**23.39%**	**26.35%**
Human Controls (Academic Prose)	88	**24.96% ± 17.47%**	*Baseline (0.00%)*	*Ref (50.00%)*	31.96%	31.23%	31.90%	32.76%	20.91%	20.49%
NVIDIA (Nemotron 3 Cohort)	84	19.13% ± 9.56%	−5.83%	41.01%	25.28%	25.54%	24.90%	25.40%	14.69%	13.73%

4.2 Domain-Slice Findings: Why Ethical Conundrums Maximize Separability

As demonstrated in Table 2, pure semantic infilling achieves an overall threshold-independent **AUROC of 68.76%** and **AUPRC of 93.21%** across the entire $N = 600$ corpus ($\Delta\mu = +14.87\%$), with remarkable variation across cognitive domains:

1. **Peak Separability in Ethical Conundrums (Slice 2: 75.76% AUROC, 94.91% AUPRC):** Synthetic prose in Slice 2 exhibits the highest mean resonance (42.93%), while authentic human ethical discourse drops to the lowest floor (21.70%), yielding a +21.23% **separation margin**. Because commercial LLMs undergo intensive RLHF safety and value alignment on moral dilemmas, they converge onto highly invariant, balanced dialectical transitions (weighing utilitarian vs. deontological trade-offs in predictable sequence). Human ethicists, conversely, advance committed, non-canonical philosophical arguments that the prober fails to anticipate.

2. **High Separability in Strategic Decisions (Slice 3: 70.67% AUROC, 93.48% AUPRC):** Strategic decision-making yields a +16.38% margin (40.69% AI vs. 24.31% Human), reflecting the canonical risk-matrix structures embedded in instruction-tuned models.

3. **Baseline Compression in Day-to-Day Applied Prose (Slice 4: 62.93% AUROC):** In routine procedural documentation, human writing naturally follows standardized templates ($\mu_{\text{Hu}} = 28.93\%$), narrowing the gap with synthetic text ($\Delta\mu = +10.40\%$).

5 Cross-Architecture Model DNA Attribution & Discursive Determinism

5.1 Multi-Component Attribution Signatures & Bidirectional Deflection

Table 3 and Table 4 reveal why continuous resonance spectroscopy captures structural provenance information that is lost in pooled binary classification:

- **DeepSeek/Qwen Achieves Highest Pairwise Separability (80.68% AUROC vs. Human):** Because DeepSeek-V3/Qwen-2.5 combines high raw bipartite reconstruction ($\bar{\mathcal{C}} = 44.37\%$, $p_4 = 46.41\%$), high Cross-Span Minimum ($S_{\min} = 27.91\%$), and low variance ($\sigma = 17.76\%$), it achieves an **80.68% pairwise AUROC** against human controls—outperforming Meta's pairwise AUROC (77.00%) despite Meta having a higher raw mean (49.30% ± 25.42%).

- **Why Pooled 6-Family Binary AUROC (68.76%) Masks Family-Specific Separability (75.69% Top-4 AUROC):** Across the four primary canonical instruction-tuned families (Meta, DeepSeek/Qwen, Google, OpenAI; $N = 344$), pooled separability against Human controls ($N = 88$)

Table 4: The 4×7 Domain Slice $\times$ Provenance Family Resonance Spectroscopy Matrix (μ_{V3} in %) on the $N = 600$ Benchmark.

Cognitive Slice	Meta (LLaMA)	DeepSeek/Qwen	Google (Gemini)	OpenAI (GPT-4o)	Anthropic (Claude)	Human Floor	NVIDIA Cohort
Slice 1: Abstract Foundations	43.05%	41.58%	38.39%	**45.14%**	30.39%	24.90%	19.17%
Slice 2: Ethical Conundrums	50.14%	45.64%	**51.33%**	49.12%	40.91%	**21.70%**	19.78%
Slice 3: Strategic Decisions	**52.33%**	44.90%	41.76%	48.42%	39.57%	24.31%	16.73%
Slice 4: Day-to-Day Applied	**51.66%**	50.47%	45.92%	29.58%	37.59%	28.93%	20.86%
All-Slice Family Mean (μ)	**49.30%**	**45.65%**	**44.35%**	**43.07%**	**37.11%**	**24.96%**	**19.13%**

reaches **75.69% AUROC** (**92.99% AUPRC**). However, when all six synthetic families ($N = 512$) are pooled into a single undifferentiated "AI" class, the aggregate AUROC moderates to 68.76% because the NVIDIA cohort ($\mu = 19.13\%$) deflects *below* the human reference floor ($\mu = 24.96\%$, directional AUROC = 41.01%). Thus, human prose occupies an intermediate baseline regime flanked by positive-deflection SFT/RLHF families (+12.15% to +24.34%) and negative-deflection high-entropy distributions (−5.83%).

- **Domain-Specific Signature Inversion (OpenAI vs. Google):** As shown in Table 4, OpenAI (GPT-4o) leads all families in *Abstract Foundations* (45.14%) and performs strongly in *Ethical Conundrums* (49.12%) and *Strategic Decisions* (48.42%), but drops to human baseline levels in *Day-to-Day Applied* prose (29.58%). Conversely, Google (Gemini 2.5) peaks in *Ethical Conundrums* (51.33%).

5.2 Deconstructing the Naive "Self-Model Resonance" Hypothesis

A striking empirical finding in Table 3 is that although our empirical reference prober is `nemotron-3-super-120b` ($M = 1$ fixed prober evaluating $G = 7$ provenance classes), the NVIDIA generative cohort yields the *lowest* resonance score ($\mu = 19.13\% \pm 9.56\%$, $S_{\min} = 14.69\%$), whereas Meta (49.30%) and DeepSeek/Qwen (45.65%) score more than double.

This directly refutes the naive assumption that an infilling prober unconditionally favors text generated by its own architectural family. Instead, we establish that **single-prober resonance spectroscopy measures the intrinsic discursive determinism of the generator**:

$$\mathbb{E}_{D \sim p_{\text{gen}}} \left[\mathcal{R}(D; P_{\text{prober}}) \right] \propto \frac{1}{H_{\text{discourse}}(p_{\text{gen}})} - D_{\text{KL}} \left(p_{\text{gen}}^{\text{macro}} \parallel p_{\text{prober}}^{\text{macro}} \right) \tag{4}$$

Models trained with strong canonical instruction-following priors (Meta LLaMA-3.3, DeepSeek-V3) collapse onto low-entropy macro-propositional trajectories ($H_{\text{discourse}} \ll 1$) that any high-capacity prober readily reconstructs. Conversely, generators utilizing high-entropy sampling or reward post-training diverged from standard SFT centroids produce lower cross-sentence infillability, yielding a negative-deflection signature (19.13% vs. 24.96% human floor).

6 Computational Profile, Evidentiary Integrity, and Limitations

To ensure strict scientific transparency and zero overstatement, we explicitly document the operational boundaries of our study:

1. **Single-Reference Prober Scope ($M = 1$) vs. Multi-Prober Generalization:** While our mathematical formulation (Section 3) defines the general M-prober resonance matrix $\mathbf{R} \in \mathbb{R}^{N \times M}$, our empirical evaluation on the $N = 600$ benchmark utilizes a single fixed 120B reference prober ($M = 1$, `nemotron-3-super-120b`) across $G = 7$ provenance classes. Expanding the active prober ensemble ($M > 1$) to compute cross-prober symmetric affinity matrices is a primary direction for future work.

2. **Compute Asymmetry and Tier-2 Forensic Positioning:** Unlike lightweight token-curvature estimators (e.g., Fast-DetectGPT or Binoculars executing on 3B–7B white-box logits in $\approx$ 50–80 ms), Pure Cloze Engine V3 queries a 120B parameter black-box prober across three void horizons ($k \in$

$\{2, 3, 4\}$), requiring 2.4–4.0 seconds per document. We explicitly position this framework as an **Asynchronous Tier-2 Forensic Audit Protocol** rather than a real-time packet filter.

3. **Moderate Pooled Separability Under Zero Surface Stylometrics:** Because we strictly forbid all surface punctuation, lexical frequency, and supervised classifier weights, pooled 6-family AUROC is moderate (68.76% overall; 75.69% across the top-4 canonical families; 80.68% on DeepSeek/Qwen). Pure cloze resonance is designed to provide interpretable, clause-by-clause bipartite matching evidence (ISO/IEC 27037 compliant) on documents of $\geq$ 6 sentences ($\geq$ 100 words), rather than black-box binary thresholding on short snippets.

7 Conclusion

We have established **Prober-Generator Resonance Spectroscopy**, a pure cloze infilling framework for characterizing LLM signature coherence without surface stylometrics or brittle binary thresholding. Evaluated across the $N = 600$ four-slice forensic benchmark, our method demonstrates how RLHF alignment and discursive determinism govern multi-sentence reconstructibility—achieving **80.68% pairwise AUROC** on DeepSeek/Qwen, **77.00% AUROC** on Meta LLaMA-3.3, and **75.76% AUROC** (**94.91% AUPRC**) in ethical reasoning domains while mapping seven provenance classes onto a continuous, mathematically grounded resonance spectrum.

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